

Prediction of pressure coefficients on roofs of low buildings using artificial neural networks

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Abstract

This paper describes an artificial neural network (ANN) approach for the prediction of mean and root-mean-square (rms) pressure coefficients on the gable roofs of low buildings. The ANN models, which employ a backpropagation training algorithm, are capable of generalizing the complex, nonlinear functional relationships between the pressure coefficients and eave height, wind direction and spatial location on the roof. The performance of the ANN is demonstrated by the prediction of the pressure coefficients for roof tap locations in a corner bay. The mean bay uplift can be predicted accurately with an average error less than 2% for three cornering wind directions not seen by the ANN during training. The mean-square errors of all of the individual pressure taps in the corner bay were 12% and 9% for the mean and rms coefficients, respectively. This approach could be used to expand aerodynamic databases to a larger variety of geometries and increase its practical feasibility.

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1. Introduction

The prediction of wind-induced pressure coefficients on the wide variety of building geometries is of considerable practical importance. Time-consuming and costly wind tunnel tests can only cover a limited number of basic configurations since pressure distributions depend on a large number of variables including incident wind

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direction, α , roof pitch, β , eave height, H , upstream terrain roughness, z_0 , building dimensions (B, L), etc. Codes of practice are generally based on a handful of pressure coefficients for a range of building geometries which are expected to envelope the worst structural and cladding loads. Simiu and co-workers (e.g., Whalen et al. [1]) have proposed making use of aerodynamic databases in order to make designs more risk-consistent, more economical and safer than those based on conventional codes. In order to be of practical use, there is a need to explore accurate techniques to interpolate or expand the aerodynamic database. In addition, vulnerability models used by (re)insurance companies are in need of accurate estimates of pressure coefficients beyond the normal set of building geometries which codes of practice usually contain. The present work focuses on the development of a predictive tool to accurately estimate pressure coefficient *statistics* on buildings for which no, or limited, data exist. This tool is based on artificial neural networks (ANNs) and makes use of the extensive aerodynamic database being developed for the NIST project (e.g., [2]).

Previously, we reported a time-delay ANN approach [3] to interpolate and predict pressure *time series*. This approach was used to predict the pressure coefficient time series at a tap location, given the time series at adjacent tap locations. This tool was developed in order to increase the resolution of pressure data being used as input for a structural analysis program without the problems associated with spatial aliasing that occur with simple linear interpolation. The main downside of this approach was the much higher level of complexity and computational effort. With this in mind, we have developed a simpler ANN approach to simulate the pressure time series necessary for structural load analyses in a companion paper [4]. The important contribution is the ability of the proposed technique to simulate/interpolate the pressure time series on a building for which no data exist, based on data from a “similar” building in the database for which data already exist. The tool involves employing an accurate interpolation technique to predict the mean and root-mean-square (rms) pressure coefficients and then generating the pressure time series. The tool is relatively simple and thus could be practically feasible for implementation within the aerodynamic database control software. The current paper discusses the technique for predicting statistics of pressure time series.

To address the prediction and interpolation issues of nonlinear multivariate modeling problems, there are many methods, such as linear interpolation, regression polynomials and ANN. Linear interpolation is simple and useful, but obviously it cannot solve nonlinear, multivariate modeling problems. Regression polynomials are the most common method for obtaining empirical equations to fit experimental data. But even for a general wind tunnel test, given the limited and sometimes inconsistent data involving a large number of variables, it is very cumbersome and difficult to develop the empirical generalization since this is a multidimensional surface fitting problem where the underlying functional forms are complex and unknown. In this case, ANN have an advantage in that multilayer perceptron neural networks are equivalent to analytical functions which can describe a multidimensional surface or a group of multidimensional curves [5]. ANN have the ability to learn and generalize the complex, nonlinear functional relationships via training with sample data

obtained from computational or experimental results, even given noisy or incomplete information [6,7], thus providing an efficient alternative solution to common interpolation schemes. The main drawback is that there is currently no theoretical basis for determining an optimum neural network configuration.

ANN have been reported in a number of engineering applications such as turbulence control for drag reduction, structural dynamic model identification and many others [8]. They also have been previously applied in the field of wind engineering. Examples include predicting the wind load distribution for air-supported structures [5], simulating wind-induced pressure time series on low buildings (point-to-point time series prediction) [9], modeling wind-induced interference effects due to adjacent buildings [10–11], and predicting wind-induced damage [12].

The objective of the present work is to develop ANN models which are accurate and robust for the prediction and interpolation of mean and rms pressure coefficients anywhere on the gable roofs for any wind direction for low buildings within the aerodynamic database. Presently, only a single building plan and roof slope are considered. Thus, given a roof height, wind direction and location coordinates on the roof, the ANN provide the corresponding mean and rms pressure coefficients.

2. Experimental data

Wind tunnel experiments were carried out in the Boundary Layer Wind Tunnel II at the University of Western Ontario. The wind tunnel is of a recirculation type and has a 39.0 m long test section, 3.4 m in width and 2.5 m in height. Surface pressure time series were acquired on a 1:100 length scale, gable-roofed building model of plan area 80 ft (24.38 m) $\times$ 125 ft (38.10 m) with a roof slope of 1 in 12. Four building heights, 16, 24, 32 and 40 ft (4.88, 7.32, 9.75 and 12.19 m), were employed. Fig. 1 shows the building dimensions and pressure tap layout on the roof surface. A total of 665 pressure taps were instrumented over the entire building surface with 335 pressure taps on the roof.

A high-speed solid-state pressure scanning system was used to take the measurements. The pressure taps were connected with the measurement system through PVC tubing. The Pressure signals were sampled at 500 Hz for 100 s and were measured essentially simultaneously. Assuming that the wind tunnel/full-scale velocity scale is equal to 1:3, the corresponding full-scale sampling frequency is 15 Hz according to the similarity requirement of the reduced frequency:

$$\left(\frac{Lf}{V}\right)_{\text{model}} = \left(\frac{Lf}{V}\right)_{\text{prototype}} \quad \left(\text{i.e., frequency scale} = \frac{\text{velocity scale}}{\text{length scale}}\right),$$

where L , V , and f represent the building dimension, wind velocity and sampling frequency, respectively. Each time series record (of 50,000 data points) is then equivalent to about 1 h in full-scale. Pressure time series in the aerodynamic database were corrected for residual non-simultaneity and were digitally low-pass filtered at 200 Hz. The measurements taken within the sampling cycle have a maximum time

dynamic pressure at a reference height above the wind tunnel boundary layer. The pressure coefficients can be easily transformed into full-scale wind pressures, given the upstream dynamic pressures at the reference height.

Pressure time series were measured for four roof heights, 37 wind directions, and two upstream terrains. The 37 wind directions were between 180° and 360° in increments of 5° , as illustrated in Fig. 1. Two target upstream terrains (open country and suburban) were modeled in the wind tunnel model tests. These matched the Exposure C (open country) and Exposure B (suburban) described in ASCE 7-98 [13]. The simulation of the boundary layer profiles for the terrains were based on the wind characteristics described by ESDU 82026(1), 83045(2) and 74031(3) [14–16] for mean wind speed profile, wind turbulence and wind spectrum. The simulated exposures have equivalent roughness lengths, z_0 , of 0.03 and 0.3 m, respectively, for open country and suburban terrain. Further details of the experiments can be obtained in [2].

3. Artificial neural networks

ANNs are inspired from the biological sciences by attempting to emulate the behaviour and complex functioning of the human brain in recognizing patterns [6]. They are composed of several layers of many interconnected neurons operating in parallel. Since ANN have many inputs (external stimuli) and outputs (responses) and allow nonlinearity in the transfer functions of the neurons, they can be used to solve multivariate and nonlinear modeling problems. Of course, ANNs are gross simplifications of the human brain function. In any case, they can be viewed as elaborate, nonlinear systems which, given an external input, will estimate the corresponding output, even if the inputs are noisy or incomplete.

3.1. Backpropagation neural networks

There are a diverse range of neural networks, such as multilayer perceptron networks, Carpenter networks and Hopfield networks [6,17]. The feed-forward multilayer networks using a backpropagation training algorithm are employed in the present work. This is probably the most widely used ANN because of its proven performance for functional approximations, optimization, and classification. The topology of this ANN is shown in Fig. 2.

Backpropagation neural networks, such as that shown in Fig. 2, generally have a layered structure with an input layer, an output layer and one or more hidden layers. Each hidden and output layer is composed of artificial neurons which are interconnected via adaptive weights. The ANN functionality is determined by inputs and weights (interconnections). The setting of weights is done through a training process with input–output data pairs. The training process of the backpropagation algorithm involves two stages: *feed-forward* and *error back-propagation*. In the *feed-forward* stage, as shown in Fig. 3, the input to a neuron is the sum of the weighted inputs and bias. This summation is then passed through a

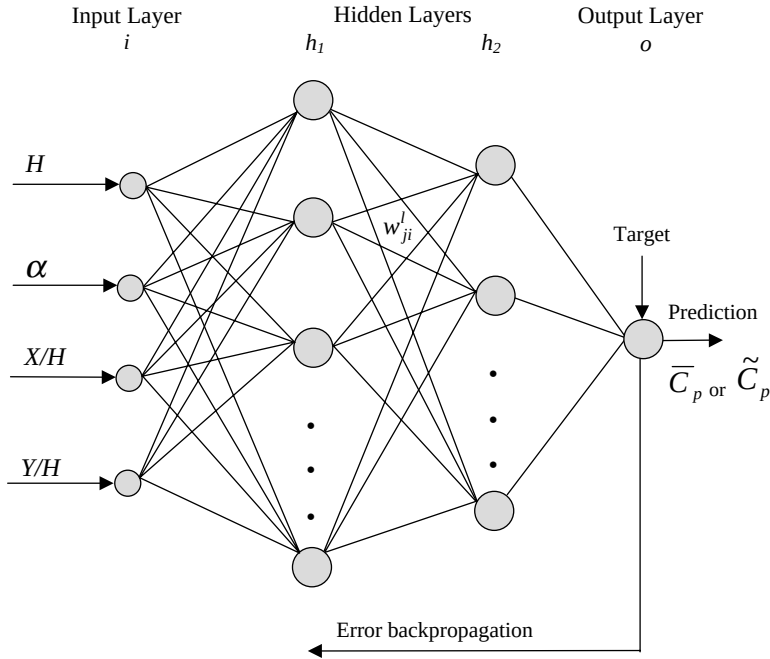


Fig. 2. Topology of the backpropagation neural network (ANN $i-h_1-h_2-o$).

nonlinear transfer function, $f(\bullet)$, thereby generating an output signal, which can be written as

$$\hat{y}_j = f\left(\sum_{i=1}^N w_{ji}x_i + b_j\right),$$

where w_{ji} is the weight that connects the i th neuron of the previous layer to the j th neuron of the current layer, x_i is the i th input and b_j is the bias associated with the j th neuron of the current layer. The transfer function, $f(\bullet)$, can be any function in principle, but the most frequently used are the nonlinear, continuous sigmoid functions. After the *feed-forward* process, the output, $\hat{y}_j$, in the output layer, may not match the desired output. The network output error (i.e., the sum of mean-square errors) is then back-propagated from the output neurons to the hidden neurons by using the backpropagation learning algorithm. The algorithm determines the weight values by minimizing the sum of mean-square errors in weight space. This *error backpropagation* process is continued until the network training has reached a stopping criterion (e.g., the network output error is less than a user-defined target). After the training process, ANNs are capable of generalizing functional relationships between input and output data patterns. In any case, these ANN functional relationships can be viewed as nonlinear functions which are dependent on inputs and coefficients (weights). Further details about the backpropagation algorithm can

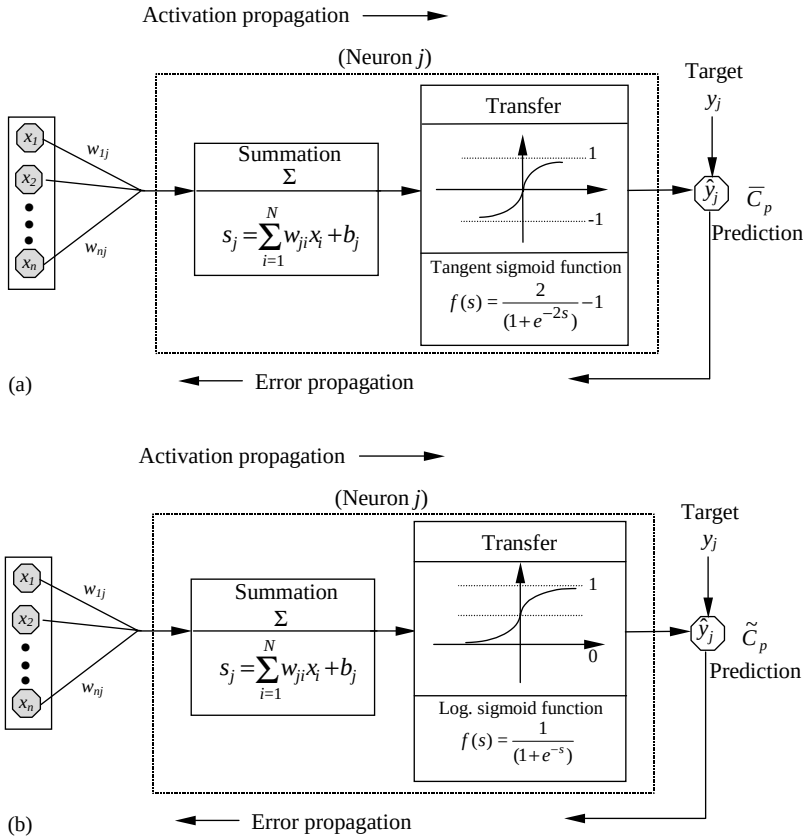


Fig. 3. A basic neuron in the hidden layers for (a) prediction of $\bar{C}_p$ with ANN 4-15-10-1, and (b) prediction of $\tilde{C}_p$ ANN 4-18-12-1.

be obtained from many sources (e.g., [6,17,18]). The Levenberg–Marquardt backpropagation training algorithm, a standard numerical optimization technique which approaches second order training speed without having to compute the Hessian matrix [19], was applied in this work because of its fast convergence in training. In order to overcome the “overfitting” (or overtraining) problem and improve the generalization, the early stopping technique (i.e., cross-validation) [17,19] was used during training.

3.2. Parameters of the employed ANN

Different learning rates, learning algorithms, transfer functions, and the numbers of neurons in the hidden layers will affect computational efficiency. Currently, there are no standard rules for choosing the proper neural network structure. There are, however, some guidelines to place an upper bound on the network size such as

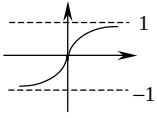
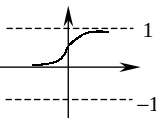
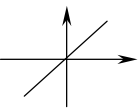
(i) using no more than three layers (excluding input layer) for the fully connected multilayer networks, and (ii) using fewer hidden layer neurons than training data [17]. Specific structures are problem dependent and are fixed by trial-and-error, especially when given little or no prior knowledge of the problem.

In order to predict mean and rms pressure coefficients, two three-layer backpropagation neural network structures were chosen since three-layer fully connected ANNs are capable of forming an arbitrarily close approximation to any continuous nonlinear mapping [6,17]. By using a trial-and-error approach which starts with a small number of hidden neurons and then increasing the size until the training performance is acceptable, three-layer network structures with varying numbers of hidden neurons and different training parameters have been investigated. For example, network structures ANN 4-12-8-1, ANN 4-14-10-1, ANN 4-15-10-1, ANN 4-18-10-1, ANN 4-18-12-1, ANN 4-20-15-1 (and others) were investigated. (Here, the notation ANN i - h_1 - h_2 - o is used as a label for a net with i input variables, o output variables and two hidden layers with h_1 and h_2 neurons in the first and second hidden layers, respectively.) The mean-square error (MSE) range was between 0.01003 and 0.00108. Among them, ANN 4-15-10-1 and ANN 4-18-12-1 were found to have the best performance to predict the mean, $\bar{C}_p$, and the rms, $\tilde{C}_p$, pressure coefficients, respectively. The transfer functions, training parameters and training performance of the developed ANN models are summarized in Table 1 and the corresponding training processes for prediction of the mean and rms pressure coefficients are shown in Fig. 4.

During training, several stopping criteria were used: (i) when the MSE increases for a specified number of consecutive iterations (e.g., five iterations) using the validation data, (ii) when the training error is less than a training goal (e.g., $1.0\text{e}-4$), and (iii) when the magnitude of the training gradient is less than a small specified value (e.g., $1.0\text{e}-10$). Whenever one of these stopping criteria is first met, training is terminated. Due to the use of the Levenberg–Marquardt backpropagation algorithm and the early stopping technique, the training error converged very quickly, as shown in Fig. 4. It took no more than 151 training iterations (about 10 min on an IBM workstation) to converge for the ANN models. It is noted that the difference between ANN 4-15-10-1 and ANN 4-18-12-1 is not only the different numbers of neurons in hidden layers, but also the different nonlinear transfer functions employed in the neurons, as shown in Fig. 3 and Table 1. For prediction of $\bar{C}_p$, the transfer function is a hyperbolic tangent sigmoid function, but for $\tilde{C}_p$ prediction, a logarithmic sigmoid transfer function was employed (since $\tilde{C}_p$ is positive while $\bar{C}_p$ can be positive and negative).

In wind tunnel tests of low buildings, there are a large number of variables involved, such as wind direction, roof pitch, eave height, building plan dimensions, and upstream terrain. Under similar flow conditions, the flow pattern around a building strongly depends on wind direction and roof pitch. Accordingly, the distribution of wind-induced pressures changes significantly with these varying parameters. In the present work, due to the current limitations of the experimental database, wind direction, α , roof height, H , and normalized roof coordinates (X/H , Y/H) were considered as input variables to the ANN. This is sufficient to clearly

Table 1
Parameters of the employed backpropagation neural networks

Neural networks	ANN 4-15-10-1 (for $\bar{C}_p$) ANN 4-18-12-1 (for $\tilde{C}_p$)
Inputs	$H, \alpha, X/H, Y/H$
Output	$\bar{C}_p$ or $\tilde{C}_p$
Transfer function	Tangent Sigmoid (Hidden layers, for $\bar{C}_p$)  Logarithmic Sigmoid (Hidden layers, for $\tilde{C}_p$)  Linear (Output layer) 
Training data	$\alpha = 270-360^\circ$ (except $300^\circ, 320^\circ$ and 340°), $H = 24, 32$ and 40 ft (7.32, 9.75 and 12.19 m), $0 < x < 125$ ft and $0 < y < 80$ ft ($0 < x < 38.1$ m and $0 < y < 24.38$ m) (a total of 335 pressure taps).
Validation data	$\alpha = 180-270^\circ, H = 24, 32$ and 40 ft (7.32, 9.75 and 12.19 m)
Test data	$\alpha = 300^\circ, 320^\circ$, and $340^\circ, H = 24, 32$ and 40 ft (7.32, 9.75 and 12.19 m)
Training terrain	Open country ($z_0 = 0.03$ m)
Training algorithm	Levenberg–Marquardt backpropagation
Learning rate	0.6
Maximum failing number	Five times (used in the validation)
Training epoch	151 epoches for $\bar{C}_p$ 122 epoches for $\tilde{C}_p$
Training performance (MSE)	0.00108 for $\bar{C}_p$ 0.00129 for $\tilde{C}_p$

illustrate the power of the technique. The output is either $\bar{C}_p$ or $\tilde{C}_p$. Once a larger aerodynamic database which consists of a wide range of building geometries is available, more input variables (e.g., roof pitch, upstream roughness length, building plan dimensions) can be conveniently incorporated into the ANN models by re-training the networks.

Considering the symmetry of the building, attention was paid to experimental data within the wind direction range from 270° to 360° . Since the early stopping technique was employed to avoid “overfitting” problem during training, the available experimental data were allocated into three subsets: training data, validation data

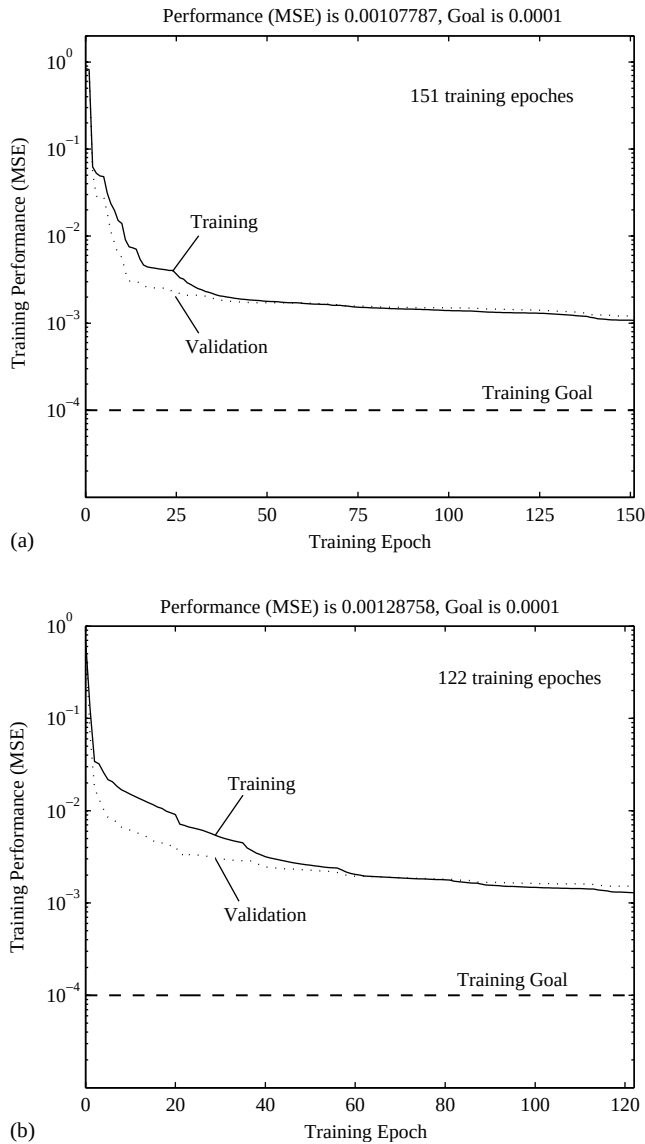


Fig. 4. Training process of (a) ANN 4-15-10-1 and (b) ANN 4-18-12-1 using the early stopping technique.

and testing data, as shown in Table 1. The training data which consist of the experimental input–output data pairs from 270° to 360° (except 300° , 320° and 340°) at three roof heights, 24, 32 and 40 ft (7.32, 9.75 and 12.19 m) in the open country terrain, were used to train the ANN functions. The experimental data pairs between 180° and 270° were used as the validation data. The purpose of using

the validation data is to monitor the training process (but not to adjust the weights) so that the ANN would not be “overtrained” and the generalized performance can be improved. As shown in Fig. 4, the output error with the validation data will normally decrease during the initial phase of training. After some training epochs, when a network begins to overfit the data, validation error would typically begin to rise. When validation error increases continuously for a specified number of iterations (called the “maximum failing number”), training is stopped and the weights for the minimum of the validation error are returned. The experimental data of three cornering wind directions (300° , 320° and 340°) were chosen as the new test data and were not used during the training or validation; these data were used for evaluating the generalized performance of the trained net.

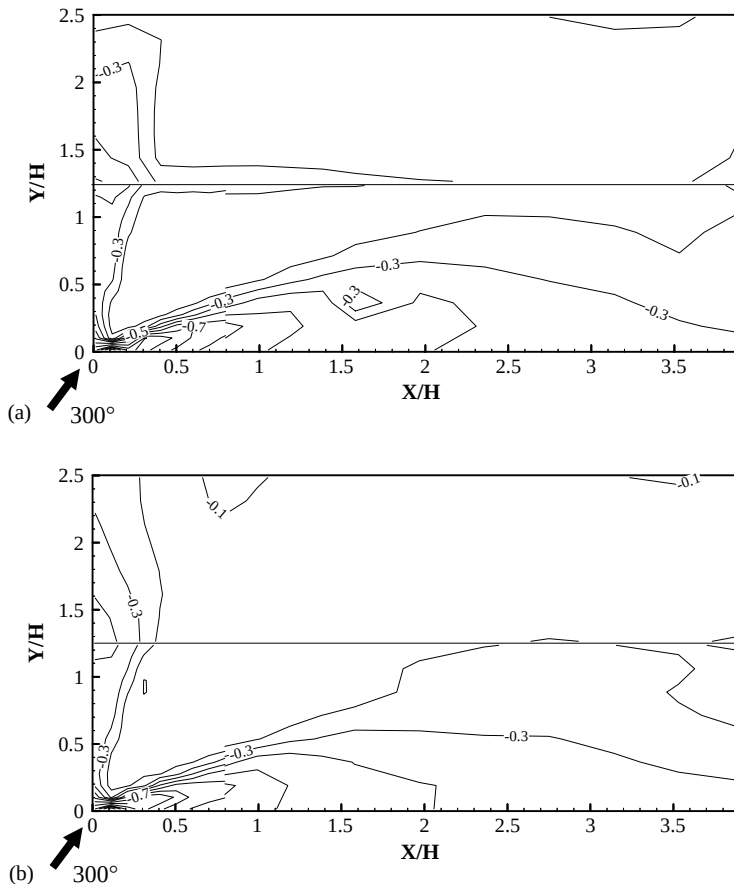


Fig. 5. Contours of $\bar{C}_p$ for a wind direction of 300° at roof height 32 ft (9.75 m) in open country terrain: (a) experimental data and (b) ANN prediction.

4. Prediction of $\tilde{C}_p$ and $\tilde{C}_p$

By training ANN with the input–output example patterns, the multivariate nonlinear functional relationships are captured and generalized. Thus, given the values of the input variables (α , H , X/H and Y/H), the corresponding output, $\tilde{C}_p$ or $\tilde{C}_p$, can be obtained (without doing further experiments). In order to evaluate the prediction performance of the ANN models and to determine whether the ANN is sufficiently generalized to be of practical use, the test data for the ANN prediction are the wind directions that did not participate in the training or validation processes, i.e., the cornering wind directions (300° , 320° and 340°). In other words, the developed ANN model will be used to interpolate or expand the current database into these three new wind directions. The prediction performance for other wind

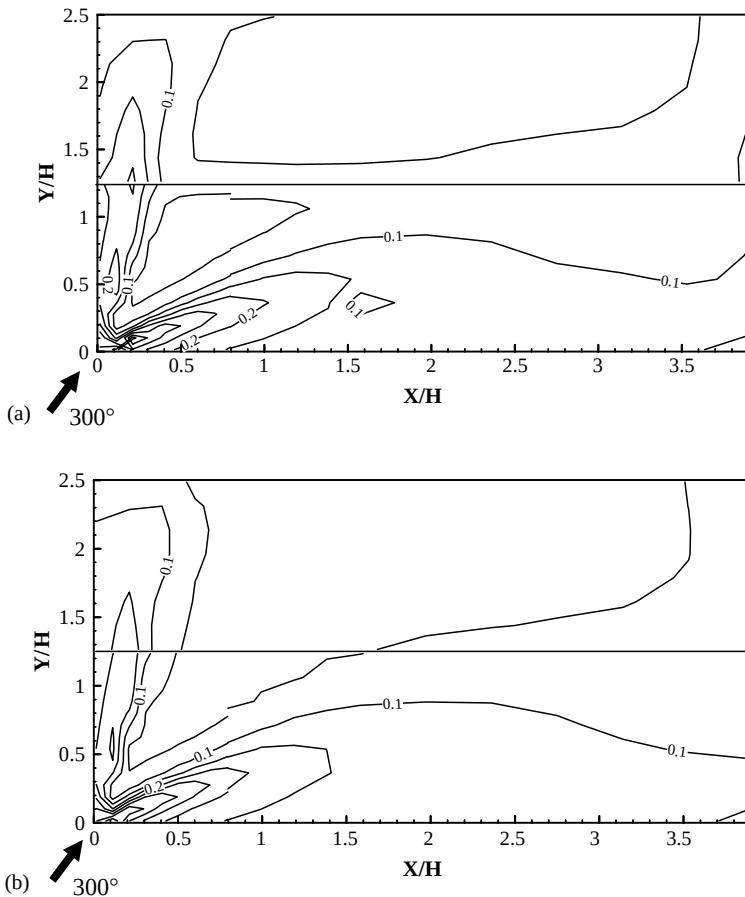


Fig. 6. Contours of $\tilde{C}_p$ for a wind direction of 300° at roof height 32 ft (9.75 m) in open country terrain: (a) experimental data and (b) ANN prediction.

directions will not be presented here in detail since these data have been used to either train the ANN models or monitor the training process (i.e., validation). The ANNs are expected to perform the prediction well when given the training and validation data, as illustrated by the corresponding error convergence in Fig. 4.

Figs. 5 and 6 show the comparison of distribution of $\bar{C}_p$ and $\tilde{C}_p$ over the roof surface between the measured data and ANN prediction for the previously “unseen” wind direction of 300° at roof height 32 ft (9.75 m). The corresponding comparison of contour plots of $\bar{C}_p$ and $\tilde{C}_p$ for the other two wind directions (320° and 340°) are depicted in Figs. 7–10, respectively. As shown in these figures (Figs. 5–10), the ANN predictions are in good agreement with the experimental data over the whole roof surface, including the leading corner and edge areas. The leading corner and edges are expected to be difficult to predict because these areas are characterized by large

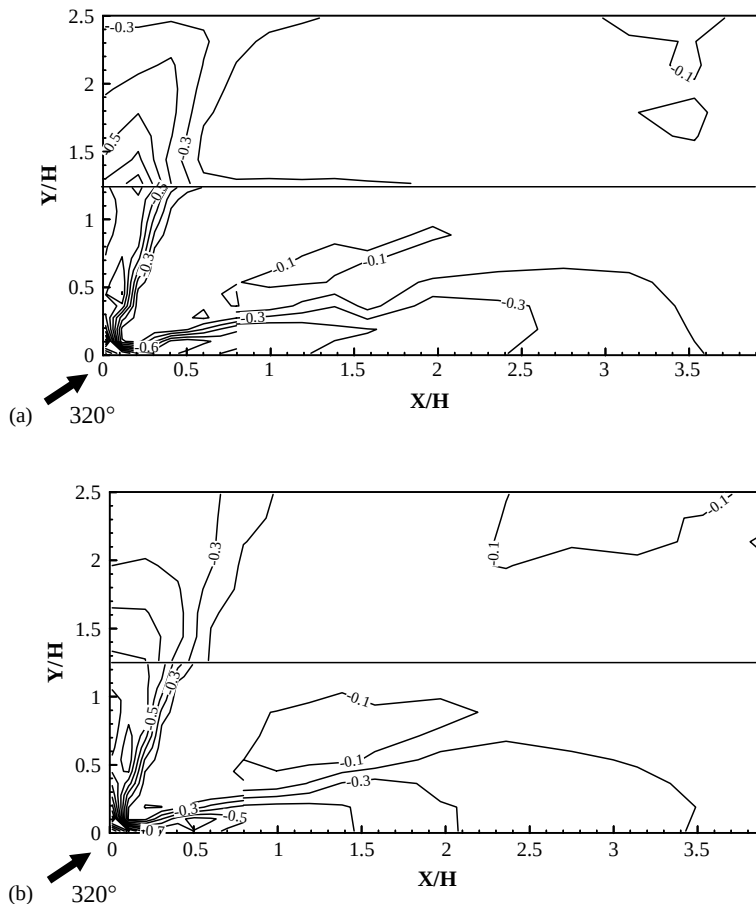


Fig. 7. Contours of $\bar{C}_p$ for a wind direction of 320° at roof height 32 ft (9.75 m) in open country terrain: (a) experimental data and (b) ANN prediction.

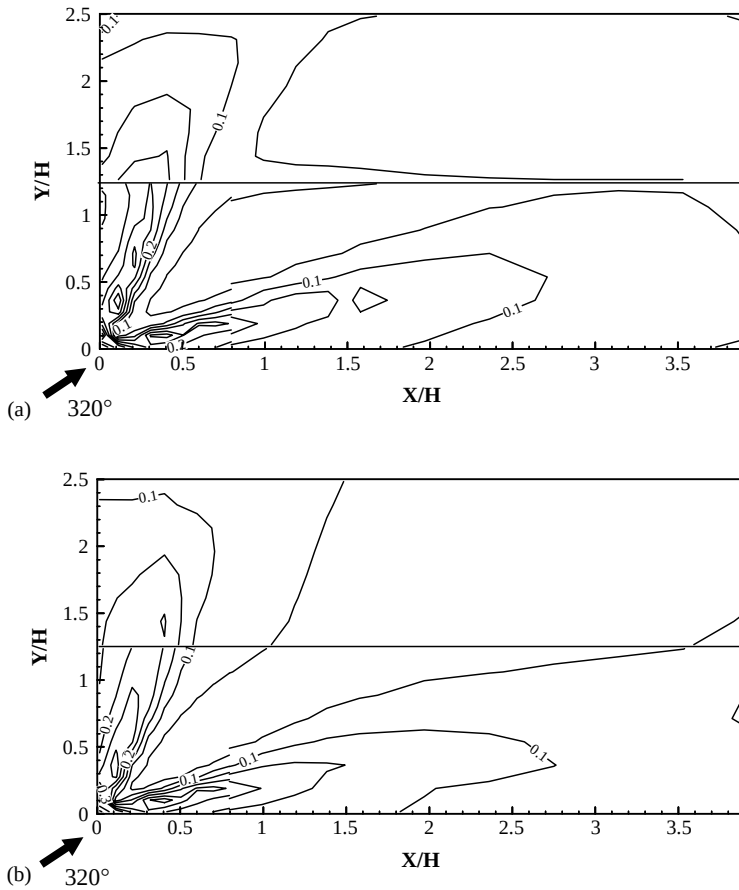


Fig. 8. Contours of $\bar{C}_p$ for a wind direction of 320° at roof height 32 ft (9.75 m) in open country terrain: (a) experimental data and (b) ANN prediction.

pressure gradients and high rms values associated with flow separation and corner (conical) vortices. The worst suctions often occur in these regions due to the strong flow separation at the leading edge. The effects of the corner vortices are clearly visible in the contour distribution of $\bar{C}_p$ and $\bar{C}_p$ predicted by the ANN. It is observed that the sharp pressure gradients in the near-corner and leading edge areas were captured by the ANN, but seem to be missed along the leeward ridge. This can be attributed to lower pressure tap resolution along the ridge than that in the corner bay zone, as shown in Fig. 1. It is also found that the ANN prediction tends to yield a smooth contour plot, which indicates a good prediction because all experimental data may contain some noisy or erratic points. A well-trained ANN will not memorize every data point, but will recognize and generalize the informative and meaningful patterns [6]. Smooth curves are observed in all the figures even though the data is plotted at the same resolution as the original data.

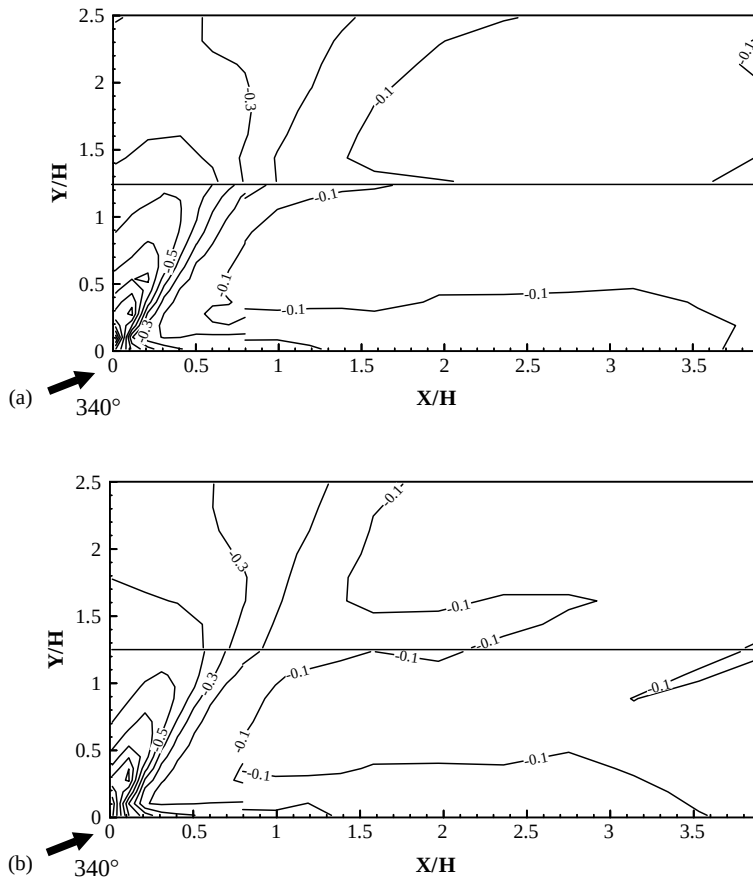


Fig. 9. Contours of $\bar{C}_p$ for a wind direction of 340° at roof height 32 ft (9.75 m) in open country terrain: (a) experimental data and (b) ANN prediction.

Since the corner area is the most difficult to be predicted, an individual corner tap (tap #702) and a leading-corner bay are taken as examples to illustrate the predictive performance in detail (see Fig. 1 for the definitions of the corner bay and corner tap). Table 2 summarizes the prediction performance of the ANN for $\bar{C}_p$ and $\bar{C}_p$ at the corner tap and corner bay for seven wind directions (270° , 290° , 300° , 320° , 325° , 340° and 360°) at roof heights 24, 32 and 40 ft (7.32, 9.75 and 12.19 m). Again, the cornering wind directions (300° , 320° and 340°) are new test directions which have not been “seen” previously by the ANN during training. As shown, the ANN achieves good performance in predicting $\bar{C}_p$ and $\bar{C}_p$ for both individual pressure taps and the mean uplift load in the corner bay. Herein, the MSE is calculated as the mean-square of the prediction errors for all 120 individual pressure taps in the corner bay.

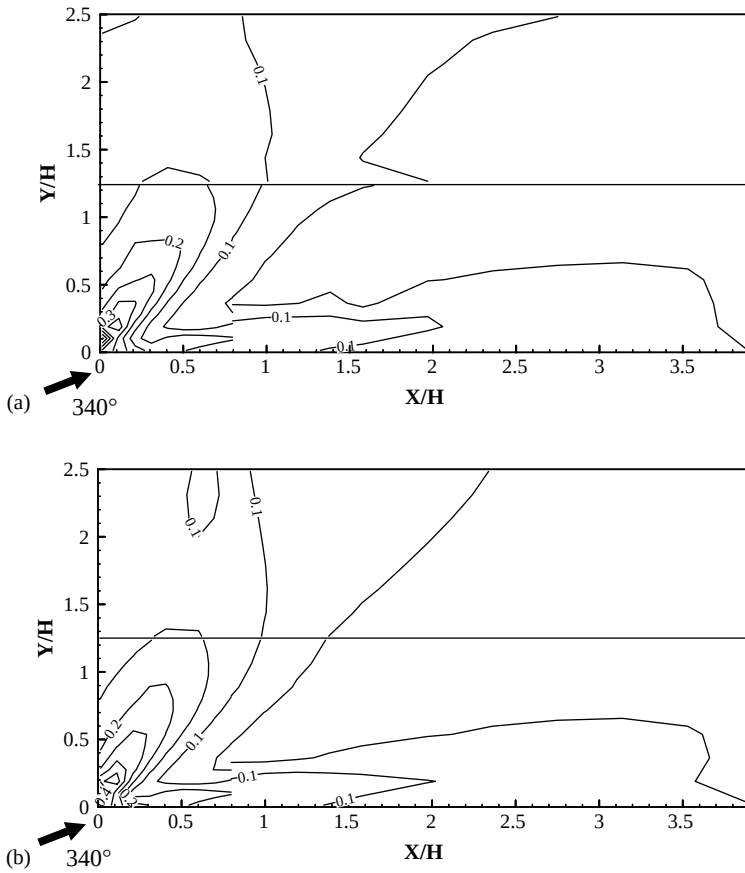


Fig. 10. Contours of $\bar{C}_p$ for a wind direction of 340° at roof height 32 ft (9.75 m) in open country terrain: (a) experimental data and (b) ANN prediction.

For the corner tap, the ANN can perform well where the mean and rms pressure coefficients can be predicted with average errors of about 7% and 5%, respectively, for all the wind directions including the three unseen cornering wind directions. The corresponding maximum absolute error of the ANN predictions for $\bar{C}_p$ and $\tilde{C}_p$ is approximately 13%. In the corner bay, the average MSE of all 120 pressure taps are less than 12% and 9% for the prediction of $\bar{C}_p$ and $\tilde{C}_p$, respectively. This indicates that the overall prediction for individual taps in the corner bay is good.

The mean bay uplift was also calculated with the individual predicted $\bar{C}_p$. It is shown that the mean uplift load of the corner bay can be predicted accurately with an average error of less than 2% for all the wind directions and roof heights. The corresponding maximum prediction error is below 6%. It is also observed that the prediction errors for the cornering wind directions tend to become relatively larger than other wind directions due to the stronger corner vortices and higher gradients. Nevertheless, it can be concluded that the ANN approach is robust when applied to

Table 2

Prediction performance of ANN for $\tilde{C}_p$ and $\tilde{C}_p$ for wind directions of 270–360°

H (ft)	α	$\tilde{C}_p$							$\tilde{C}_p$			
		Corner tap			Bay mean uplift			MSE (%)	Corner tap			MSE (%)
		Real	ANN	Error (%)	Real	ANN	Error (%)		Real	ANN	Error (%)	
24	270°	−0.601	−0.603	0.3	−0.259	−0.268	3.5	13.0	0.236	0.239	1.3	8.5
	290°	−0.875	−0.783	−10.5	−0.233	−0.236	1.3	13.6	0.320	0.320	0.0	9.7
	300°	−0.751	−0.681	−9.3	−0.238	−0.241	1.3	15.6	0.308	0.289	−6.2	10.3
	320°	−0.540	−0.506	−6.3	−0.271	−0.281	3.7	16.7	0.212	0.230	8.5	9.3
	320°	−0.568	−0.528	−7.0	−0.278	−0.293	5.4	16.8	0.225	0.236	4.9	9.1
	340°	−0.758	−0.860	13.5	−0.307	−0.309	0.7	13.1	0.299	0.331	10.7	9.0
	360°	−0.468	−0.460	−1.7	−0.308	−0.303	−1.6	7.4	0.218	0.200	−8.3	6.3
32	270°	−0.704	−0.647	−8.1	−0.332	−0.319	−3.9	9.9	0.279	0.279	0.0	9.7
	290°	−0.994	−0.987	−0.7	−0.276	−0.281	1.8	12.7	0.347	0.369	6.3	10.8
	300°	−0.877	−0.902	2.9	−0.293	−0.289	−1.4	12.8	0.352	0.346	−1.7	11.3
	320°	−0.691	−0.612	−11.4	−0.340	−0.337	−0.9	11.8	0.254	0.287	13.0	9.1
	325°	−0.728	−0.630	13.5	−0.345	−0.349	1.2	14.4	0.273	0.290	6.2	9.8
	340°	−0.908	−1.012	11.5	−0.370	−0.363	−1.9	9.3	0.341	0.366	7.3	5.3
	360°	−0.489	−0.482	1.4	−0.356	−0.352	−1.1	4.1	0.224	0.221	−1.3	4.6
40	270°	−0.718	−0.658	−8.4	−0.370	−0.372	0.5	7.3	0.284	0.304	7.0	7.0
	290°	−1.110	−1.168	5.2	−0.332	−0.335	0.9	7.8	0.378	0.401	6.1	10.9
	300°	−1.062	−1.088	2.4	−0.354	−0.348	−1.7	10.9	0.410	0.378	−7.8	9.7
	320°	−0.824	−0.741	−10.1	−0.398	−0.399	0.3	16.0	0.292	0.317	8.6	11.8
	325°	−0.878	−0.807	−8.1	−0.406	−0.408	−0.5	15.5	0.310	0.319	2.9	9.6
	340°	−1.023	−1.118	9.3	−0.414	−0.419	1.2	11.1	0.384	0.387	0.8	9.5
	360°	−0.490	−0.518	5.7	−0.398	−0.398	0.0	3.9	0.226	0.230	1.8	5.2
Average error				7.0			1.7	11.6			5.3	8.9

Note: 1. Wind directions (300°, 320° and 340°) are test data which were not used for training or validation.
2.

$$\text{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{C_{P_{i,\text{test}}} - C_{P_{i,\text{real}}}}{C_{P_{i,\text{real}}}} \right)^2} = \text{mean square error}$$

(calculated for the corner bay).

predict the distribution of the mean and rms pressure coefficients for the different wind directions and roof heights.

5. Conclusions

The application of artificial neural networks for the prediction of pressure statistics ($\tilde{C}_p$ and $\tilde{C}_p$) on low, gable-roofed buildings was observed to be successful.

The ANN were trained with wind tunnel experimental data from generic buildings of similar plan dimensions but varying roof height. The promising results indicate the ANNs are capable of generalizing functional relationships involving a large number of variables, such as wind direction, roof height and normalized roof coordinates (X/H , Y/H) in order to predict $\tilde{C}_p$ and $\tilde{C}_p$ with good accuracy for any combination of these variables. It can be concluded that this approach has the potential to expand the experimental database to increase its practical feasibility, provided that the ANN (used to predict $\tilde{C}_p$ and $\tilde{C}_p$) have covered the broad range of aerodynamics found for low buildings.

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